SEONGKU KANG

Assistant Professor, Department of Computer Science and Engineering, Korea University
Email: seongkukang@korea.ac.kr | Google Scholar | Web Page | Lab Page

RESEARCH INTERESTS

- Data science and machine learning for real-world applications
- Information retrieval and recommender systems

EXPERIENCE

Assistant Professor, Korea University Mar 2025 - Present

- Department of Computer Science and Engineering

Postdoctoral Researcher, UIUC Aug 2023 - Feb 2025

- Data Mining Group (advisor: Prof. Jiawei Han)

Research intern, Microsoft Research Asia Oct 2022 - April 2023

- Social Computing group (mentors: Dr. Xing Xie, Dr. Jianxun Lian)
  - Award of excellence (Stars of Tomorrow, top 10%)

EDUCATION

Ph.D., Computer Science and Engineering, POSTECH Mar 2018 - Aug 2023

- Data Intelligence Lab (advisor: Prof. Hwanjo Yu)

B.S., Computer Science and Engineering, NTU Jul 2016 - May 2017

- Exchange Student

B.S., Computer Science and Engineering, Hanyang University Mar 2012 - Feb 2018

- Cumulative GPA: 4.43/4.50 (ranked 1st in CSE Dept.)

PUBLICATIONS

Unless otherwise specified, the paper is accepted as a long/regular paper. * indicates equal contribution.

Conferences

1. hoBIT: A Profile-Aware Retrieval-Augmented Chatbot for University Academic Advising
   Kim Yoonseo, Seongmin Lee, Joongheon Kim, SeongKu Kang
   EMNLP'26 Demo | Conference on Empirical Methods in Natural Language Processing
2. PEARL: Front-Loading Relational Chains for Multi-Hop Table Retrieval
   Subeen Ho, Hyeongu Kang, SeongKu Kang, Susik Yoon
   EMNLP'26 | Conference on Empirical Methods in Natural Language Processing
3. Beyond Observed Auxiliary Relations: Environment-Conditioned Modeling for Multi-Behavior Recommendation
   Seunghan Lee, Hyunsik Yoo, Jian Kang, Susik Yoon, SeongKu Kang
   CIKM'26 | ACM International Conference on Information and Knowledge Management
4. SCoRD: Semantic-Assisted Continual Retriever-Reranker Distillation for LLM-Based Recommendation
   Seunghyun Baek*, Gyuseok Lee*, Seunghan Lee, Wonbin Kweon, Dong Wang, SeongKu Kang
   CIKM'26 | ACM International Conference on Information and Knowledge Management

5. Profiling What Matters: Context-Aware Item Profiles from Large-Scale Metadata for LLM Recommenders
   Dojun Hwang, Seunghan Lee, Cheonyoung Park, Sara Yu, SeongKu Kang
   CIKM'26 | ACM International Conference on Information and Knowledge Management

6. Dynamic Spectral Decoupling for Time-Series Forecasting
   Jaehyung Lee, Seungha Hong, Seongje Lee, Suyeon Kim, SeongKu Kang†, Hwanjo Yu†
   CIKM'26 | ACM International Conference on Information and Knowledge Management

7. TRACER: Balancing Stability-Plasticity-Cognitivity Trilemma for LLM Enhanced Continual Recommendation
   Woojoo Kim, Hyunsik Yoo, Junyoung Kim, Jaehyung Lim, SeongKu Kang†, Hwanjo Yu†
   CIKM'26 | ACM International Conference on Information and Knowledge Management

8. Concepts Complement Dense Semantics: Learning Compact Sparse Spaces for Text-Image Retrieval
   Yoonseo Kim*, Jungwoo Choi*, Cheonyoung Park, Youngwook Kim, Yongho Song, SeongKu Kang
   CIKM'26 KEIR Workshop (oral) | ACM International Conference on Information and Knowledge Management

9. Uncertainty-Aware Variational Reward Factorization via Probabilistic Preference Bases for LLM Personalization
   Gyuseok Lee, Wonbin Kweon, Zhenrui Yue, SeongKu Kang, Jiawei Han, Dong Wang
   COLM'26 | Conference on Language Modeling, 2026

10. CoRank: LLM-Based Compact Reranking with Document Features for Scientific Retrieval
    Runchu Tian, Xueqiang Xu, Bowen Jin, SeongKu Kang, Jiawei Han
    KDD'26 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining

11. SPRINT: Scalable and Predictive Intent Refinement for LLM-Enhanced Session-based Recommendation
    Gyuseok Lee, Wonbin Kweon, Z Yue, Y Liu, Y Liu, Susik Yoon, Dong Wang, SeongKu Kang
    SIGIR'26 | ACM SIGIR Conference on Research and Development in Information Retrieval

12. Filling the Gaps: Selective Knowledge Augmentation for LLM Recommenders
    Jaehyun Lee, Sanghwan Jang, SeongKu Kang†, Hwanjo Yu†
    SIGIR'26 | ACM SIGIR Conference on Research and Development in Information Retrieval

13. FLAME: Condensing Ensemble Diversity into a Single Network for Efficient Sequential Recommendation
    WooJoo Kim, Junyoung Kim, Jaehyung Lim, Seong Jin Choi, SeongKu Kang†, Hwanjo Yu†
    SIGIR'26 | ACM SIGIR Conference on Research and Development in Information Retrieval

14. MVIGER: Multi-View Variational Integration of Complementary Knowledge for Generative Recommender
    Tongyoung Kim, Soojin Yoon, SeongKu Kang, Jinyoung Yeo, Dongha Lee
    SIGIR'26 | ACM SIGIR Conference on Research and Development in Information Retrieval

15. Why These Documents? Explainable Generative Retrieval with Hierarchical Category Paths
    Sangam Lee, Ryang Heo, SeongKu Kang, Susik Yoon, Jinyoung Yeo, Dongha Lee
    ACL'26 Findings | Annual Meeting of the Association for Computational Linguistics

16. Continual Low-Rank Adapters for LLM-based Generative Recommender Systems
    Hyunsik Yoo, Ting-Wei Li, SeongKu Kang, Zhining Liu, Charlie Xu, Qilin Qi, Hanghang Tong
    ICLR'26 | International Conference on Learning Representations

17. Leveraging Historical and Current Interests for Continual Sequential Recommendation
    Gyuseok Lee, Hyunsik Yoo, Junyoung Hwang, SeongKu Kang†, Hwanjo Yu
    WSDM'26 | ACM International Conference on Web Search and Data Mining

18. PairSem: LLM-Guided Pairwise Semantic Matching for Scientific Document Retrieval
Wonbin Kweon, Runchu Tian, SeongKu Kang, Pengcheng Jiang, Zhiyong Lu, Jiawei Han, Hwanjo Yu
WWW'26 | ACM The Web Conference

19. CREAM: Continual Retrieval on Dynamic Streaming Corpora with Adaptive Soft Memory
HuiJeong Son, Hyeongu Kang, Sunho Kim, Subeen Ho, SeongKu Kang, Dongha Lee, Susik Yoon
KDD'26 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining

20. Topic Coverage-based Demonstration Retrieval for In-Context Learning
Wonbin Kweon, SeongKu Kang, Runchu Tian, Pengcheng Jiang, Jiawei Han, Hwanjo Yu
EMNLP'25 | Conference on Empirical Methods in Natural Language Processing

21. Scientific Paper Retrieval with LLM-Guided Semantic-Based Ranking
Yunyi Zhang, Ruozhen Yang, Siqi Jiao, SeongKu Kang, Jiawei Han
EMNLP'25 Findings | Conference on Empirical Methods in Natural Language Processing

22. DeepRetrieval: Hacking Real Search Engines and Retrievers with Large Language Models via Reinforcement Learning
Pengcheng Jiang, Jiacheng Lin, Lang Cao, Runchu Tian, SeongKu Kang, Zifeng Wang, Jimeng Sun, Jiawei Han
COLM'25 | Conference on Language Modeling

23. Imagine All The Relevance: Scenario-Profiled Indexing with Knowledge Expansion for Dense Retrieval
Sangam Lee, Ryang Heo, SeongKu Kang, Dongha Lee
COLM'25 | Conference on Language Modeling

24. Delving into Instance-Dependent Label Noise in Graph Data: A Comprehensive Study and Benchmark
Suyeon Kim, SeongKu Kang†, Dongwoo Kim, Jungseul Ok, Hwanjo Yu†
KDD'25 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining

25. Embracing Plasticity: Balancing Stability and Plasticity in Continual Recommender Systems
Hyunsik Yoo, SeongKu Kang, Ruizhong Qiu, Charlie Xu, Fei Wang and Hanghang Tong
SIGIR'25 | ACM SIGIR Conference on Research and Development in Information Retrieval

26. Personalized Preference Reasoning with Large Language Models for Accurate and Explainable Recommendation
Jieyong Kim, Hyunseo Kim, Hyunjin Cho, SeongKu Kang, Buru Chang, Jinyoung Yeo and Dongha Lee
SIGIR'25 | ACM SIGIR Conference on Research and Development in Information Retrieval

27. Uncertainty Quantification and Decomposition for LLM-based Recommendation
Wonbin Kweon, Sanghwan Jang, SeongKu Kang†, Hwanjo Yu†
WWW'25 | ACM The Web Conference

28. Chain-of-Factors Paper-Reviewer Matching
Yu Zhang, Yanzhen Shen, SeongKu Kang, Xiusi Chen, Bowen Jin, Jiawei Han
WWW'25 | ACM The Web Conference

29. Improving Scientific Document Retrieval with Concept Coverage-based Query Set Generation
SeongKu Kang, Bowen Jin, Wonbin Kweon, Yu Zhang, Dongha Lee, Jiawei Han, Hwanjo Yu
WSDM'25 | ACM International Conference on Web Search and Data Mining

30. Unsupervised Robust Cross-Lingual Entity Alignment via Neighbor Triple Matching with Entity and Relation Texts
Soojin Yoon, Sungho Ko, Tongyoung Kim, SeongKu Kang, Jinyoung Yeo, Dongha Lee
WSDM'25 | ACM International Conference on Web Search and Data Mining

31. Taxonomy-guided Semantic Indexing for Academic Paper Search
SeongKu Kang, Yunyi Zhang, Pengcheng Jiang, Dongha Lee, Jiawei Han, Hwanjo Yu
EMNLP'24 | Conference on Empirical Methods in Natural Language Processing

32. Continual Collaborative Distillation for Recommender System
Gyuseok Lee*, SeongKu Kang*, Wonbin Kweon, Hwanjo Yu
KDD'24 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining

33. PEARL: A Review-driven Persona-Knowledge grounded Conversational Recommendation Dataset
Minjin Kim*, Minju Kim*, Hana Kim, Beong-woo Kwak, Soyeon Jeon, Hyunseo Kim, SeongKu Kang, Youngjae Yu, Jinyoung Yeo, Dongha Lee
ACL'24 Findings | Annual Meeting of the Association for Computational Linguistics

34. Self-Consistent Reasoning-based Aspect-Sentiment Quad Prediction with Extract-Then-Assign Strategy
Jieyong Kim*, Ryang Heo*, Yongsik Seo, SeongKu Kang, Jinyoung Yeo, Dongha Lee
ACL'24-short Findings | Annual Meeting of the Association for Computational Linguistics

35. Multi-Domain Sequential Recommendation via Domain Space Learning
Junyoung Hwang, Hyunjun Ju, SeongKu Kang, Sanghwan Jang, Hwanjo Yu
SIGIR'24 | ACM SIGIR Conference on Research and Development in Information Retrieval

36. Learning Discriminative Dynamics with Label Corruption for Noisy Label Detection
Suyeon Kim, Dongha Lee, SeongKu Kang, Sukang Chae, Sanghwan Jang, Hwanjo Yu
CVPR'24 | IEEE / CVF Computer Vision and Pattern Recognition

37. Improving Retrieval in Theme-specific Applications using a Corpus Topical Taxonomy
SeongKu Kang, Shivam Agarwal, Bowen Jin, Dongha Lee, Hwanjo Yu, Jiawei Han
WWW'24 | ACM The Web Conference

38. Top-Personalized-K Recommendation
Wonbin Kweon, SeongKu Kang, Sanghwan Jang, Hwanjo Yu
WWW'24 | ACM The Web Conference

39. Multi-Domain Recommendation to Attract Users via Domain Preference Modeling
Hyunjun Ju, SeongKu Kang, Dongha Lee, Junyoung Hwang, Sanghwan Jang, Hwanjo Yu
AAAI'24 | AAAI Conference on Artificial Intelligence

40. MvFS: Multi-view Feature Selection for Recommender System
Youngjune Lee, Yeongjong Jeong, Keunchan Park, SeongKu Kang†
CIKM'23-short | ACM International Conference on Information and Knowledge Management

41. Distillation from Heterogeneous Models for Top-K Recommendation
SeongKu Kang, Wonbin Kweon, Dongha Lee, Jianxun Lian, Xing Xie, Hwanjo Yu
WWW'23 | ACM The Web Conference

42. Learning Topology-Specific Experts for Molecular Property Prediction
Suyeon Kim, Dongha Lee, SeongKu Kang, Seonghyeon Lee, Hwanjo Yu
AAAI'23 | AAAI Conference on Artificial Intelligence

43. Consensus Learning from Heterogeneous Objectives for One-Class Collaborative Filtering
SeongKu Kang, Dongha Lee, Wonbin Kweon, Junyoung Hwang, Hwanjo Yu
WWW'22 | ACM The Web Conference

44. TaxoCom: Topic Taxonomy Completion with Hierarchical Discovery of Novel Topic Clusters
Dongha Lee, Jiaming Shen, SeongKu Kang, Susik Yoon, Jiawei Han, Hwanjo Yu
WWW'22 | ACM The Web Conference

45. Obtaining Calibrated Probabilities with Personalized Ranking Models
Wonbin Kweon, SeongKu Kang, Hwanjo Yu
AAAI'22 | AAAI Conference on Artificial Intelligence

46. Topology Distillation for Recommender System
SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu
KDD'21 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining

47. Bootstrapping User and Item Representations for One-Class Collaborative Filtering
Dongha Lee, SeongKu Kang, Hyunjun Ju, Chanyoung Park, Hwanjo Yu
SIGIR'21 | ACM SIGIR Conference on Research and Development in Information Retrieval

48. Unsupervised Proxy Selection for Session-based Recommender Systems
Junsu Cho, SeongKu Kang, Dongmin Hyun, Hwanjo Yu
SIGIR'21 | ACM SIGIR Conference on Research and Development in Information Retrieval

49. Learning Heterogeneous Temporal Patterns of User Preference for Timely Recommendation
Junsu Cho, Dongmin Hyun, SeongKu Kang, Hwanjo Yu
WWW'21 | ACM International World-Wide Web Conference

50. Bidirectional Distillation for Top-K Recommender System
Wonbin Kweon, SeongKu Kang, Hwanjo Yu
WWW'21 | ACM International World-Wide Web Conference

51. DE-RRD: A Knowledge Distillation Framework for Recommender System
SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu
CIKM'20 | ACM International Conference on Information and Knowledge Management

52. Deep Rating Elicitation for New Users in Collaborative Filtering
Wonbin Kweon, SeongKu Kang, Junyoung Hwang, Hwanjo Yu
WWW'20-short | ACM International World-Wide Web Conference

53. Multi-Modal Component Embedding for Fake News Detection
SeongKu Kang, Junyoung Hwang, Hwanjo Yu
IMCOM'20 | IEEE International Conf. Ubiquitous Information Management and Communication

54. Semi-Supervised Learning for Cross-Domain Recommendation to Cold-start Users
SeongKu Kang, Junyoung Hwang, Dongha Lee, Hwanjo Yu
CIKM'19 | ACM International Conference on Information and Knowledge Management
- *ranked 13th among the most influential papers at CIKM 2019*(link)

55. Densifying a Trust Network for Effective Collaborative Filtering
SeongKu Kang, Jemin Wang, Yeon-Chang Lee, Sang-Wook Kim
KDBC'17 | Korean DataBase Conference

Journals

1. Improving Scientific Document Retrieval with Academic Concept Index
Jeyun Lee, Junhyoung Lee, Wonbin Kweon, Bowen Jin, Yu Zhang, Susik Yoon, Dongha Lee, Hwanjo Yu, Jiawei Han, SeongKu Kang
ACM Transactions on Intelligent Systems and Technology (SCIE) (To appear)

2. BPL: Bias-adaptive Preference Distillation Learning for Recommender System
SeongKu Kang, Jianxun Lian, Dongha Lee, Wonbin Kweon, Sanghwan Jang, Jaehyun Lee, Jindong Wang, Xing Xie, Hwanjo Yu

IEEE Transactions on Knowledge and Data Engineering (SCIE) (2026.01)

3. Unbiased, Effective, and Efficient Distillation from Heterogeneous Models for Recommender Systems
   SeongKu Kang, Wonbin Kweon, Dongha Lee, Jianxun Lian, Xing Xie, Hwanjo Yu
   ACM Transactions on Recommender Systems (2024.02)
4. Mitigating Viewpoint Sensitivity of Self-Supervised One-Class Classifiers
   Hyunjun Ju, Dongha Lee, SeongKu Kang, Hwanjo Yu
   Information Sciences (SCIE) (2022.09)
5. Personalized Knowledge Distillation for Recommender System
   SeongKu Kang, Dongha Lee, Wonbin Kweon, Hwanjo Yu
   Knowledge-Based Systems (SCIE) (2022.03)
6. Item-side Ranking Regularized Distillation for Recommender System
   SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu
   Information Sciences (SCIE) (2021.11)

AWARDS & FELLOWSHIPS

Awards

- Seoktop Lecture Award (Bigdata Analytics), Korea University, 2026
- Seoktop Lecture Award (Artificial Intelligence), Korea University, 2025
- Top-10 Paper Recognition, WSDM 2025 / ACM TIST Best Papers of WSDM 2025
- Outstanding reviewer, ACM KDD, 2024
- Stars of Tomorrow Award, Microsoft Research Asia, 2023
- POSTECH Research Award (4th place), National Research Foundation and POSTECH, Korea, 2022
- Music Playlist Recommendation Competition (3rd place), Kakao, 2020
- SIGIR Student Travel Award, ACM SIGIR, 2019-2020
- Dean's List, Hanyang University, 2018
- Best Paper Award (3rd place), Korean Database Conference (KDBC), 2017

Fellowships

- POSTECHIAN Fellowship, POSTECH, 2023
- NAVER Ph.D Fellowship, NAVER, 2021
- NAVER Academic Scholarship, NAVER, 2017
- National Science and Technology Scholarship, Ministry of Education, Korea, 2012-2017
- Hanyang Brain Scholarship, Hanyang University, 2012

PROFESSIONAL SERVICES

Area Chair/Senior Program Committee

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD): 2026 - 2027
- System Demonstration Track, Conference on Empirical Methods in Natural Language Processing (EMNLP): 2026
- AAAI Conference on Artificial Intelligence (AAAI): 2027
- ACM International World-Wide Web Conference (WWW): 2027

Program Committee/Reviewer

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD): 2023 - 2026
- ACM International World-Wide Web Conference (WWW): 2024, 2025
- AAAI Conference on Artificial Intelligence (AAAI): 2024 - 2026
- ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR): 2025, 2026

- Annual Meeting of the Association for Computational Linguistics (ACL): 2025
- SIAM Conference on Data Mining (SDM): 2024
- Computer Vision and Pattern Recognition (CVPR): 2026
- Advances in Neural Information Processing Systems (NeurIPS): 2026
- ACM International Conference on Information and Knowledge Management (CIKM): 2026
- IEEE International Conference on Data Mining (ICDM): 2026
- ACM AI Summit: 2026
- International Conference on Learning Representations (ICLR): 2027

Journal Reviewer

- Transactions on Information Systems: 2024, 2026
- Transactions on Knowledge and Data Engineering: 2024, 2025
- Knowledge-Based Systems: 2023, 2024
- Neural Computing and Applications: 2022

Others

- Tutorial Organizer, DASFAA 2026
- Tutorial Organizer, CIKM 2025
- PhD Mentor, CIKM 2025 PhD Symposium

TALKS

- Invited Talk: Hierarchical Knowledge Indexing for Scientific AI, Samsung SAIT — Oct 2026
- Invited Seminar, Data Intelligence Workshop — May 2026
- CSE/GSAI Seminar, Pohang University of Science and Technology (POSTECH) — Mar 2026
- Tutorial: Continual Recommender Systems, ACM CIKM — Nov 2025
- Panel discussion, Data Intelligence Workshop — Jun 2025
- CS/AI Seminar, Ulsan National Institute of Science and Technology (UNIST) — Nov 2023
- Invited Seminar, Electronic & Information Research Information Center (EIRIC) — Sep 2023
- Invited Seminar, Social Computing Group, Microsoft Research Asia (MSRA) — Jun 2023
- Technology Showcase, Korea AI summit — Dec 2022